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M.Sc. Sem-2
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EXPERIMENT NO:10 **DATE: / /**

AIM : GAS DETECTION USING MQ135/MQ2 AND ESP32.

OBJECTIVE: The main objective of this experiment is,

1. To interface Gas Detecting Sensor with ESP32.

APPRATUS: ESP32, Arduino IDE, Breadboard, Resistors, Gas Detecting sensor MQ-135/MQ-2, connecting wires, supply source, Display.

DESCRIPTION:

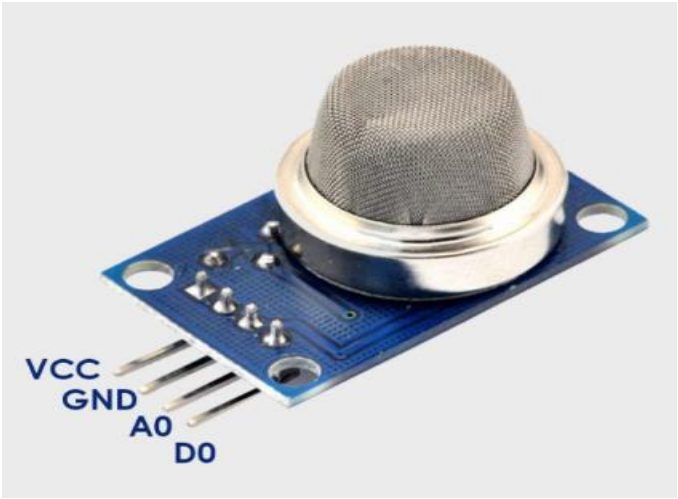
The MQ-2 sensor is a versatile gas sensor capable of detecting a wide range of gases including alcohol, carbon monoxide, hydrogen, isobutene, liquefied petroleum gas, methane, propane, and smoke. It is popular among beginners due to its low cost and easy-to-use features.

The MQ-2 sensor works on the principle of resistance changes in the presence of different gases. When the target gas comes in contact with the heated MOS (Metal Oxide Semiconductor) material, it undergoes oxidation or reduction reactions that change the resistance of the MOS material. It is noteworthy that the MQ2 gas sensor is capable of detecting multiple gases, but lacks the ability to differentiate between them. This is a common characteristic of most gas sensors. The sensor has a built-in potentiometer that allows you to adjust the sensor digital output (D0) threshold. When the concentration of gas in the air exceeds a certain threshold value, the resistance of the sensor changes.The core of the sensor is a ceramic tube coated with a thin layer of SnO_2. In clean air, SnO_2 has a high electrical resistance. This is because oxygen molecules from the atmosphere are adsorbed onto the surface of the SnO_2 layer and "trap" free electrons, creating a potential barrier that prevents current flow. In a metal mesh cover, a small heating filament is present. It is essential because the chemical reactions required for detection only occur efficiently at high temperatures (typically

around **300°C**). This is why the sensor needs a **pre-heating period** and becomes physically hot to the touch during operation.The gas molecules react with the adsorbed oxygen ions.This reaction releases the trapped electrons back into the SnO_2 layer.As free electrons increase, the electrical resistance of the sensor drops.The sensor module uses a simple **voltage divider** circuit with a load resistor RL.

- **Low Gas Concentration:** High sensor resistance = **Low output voltage**.
- **High Gas Concentration:** Low sensor resistance = **High output voltage**.

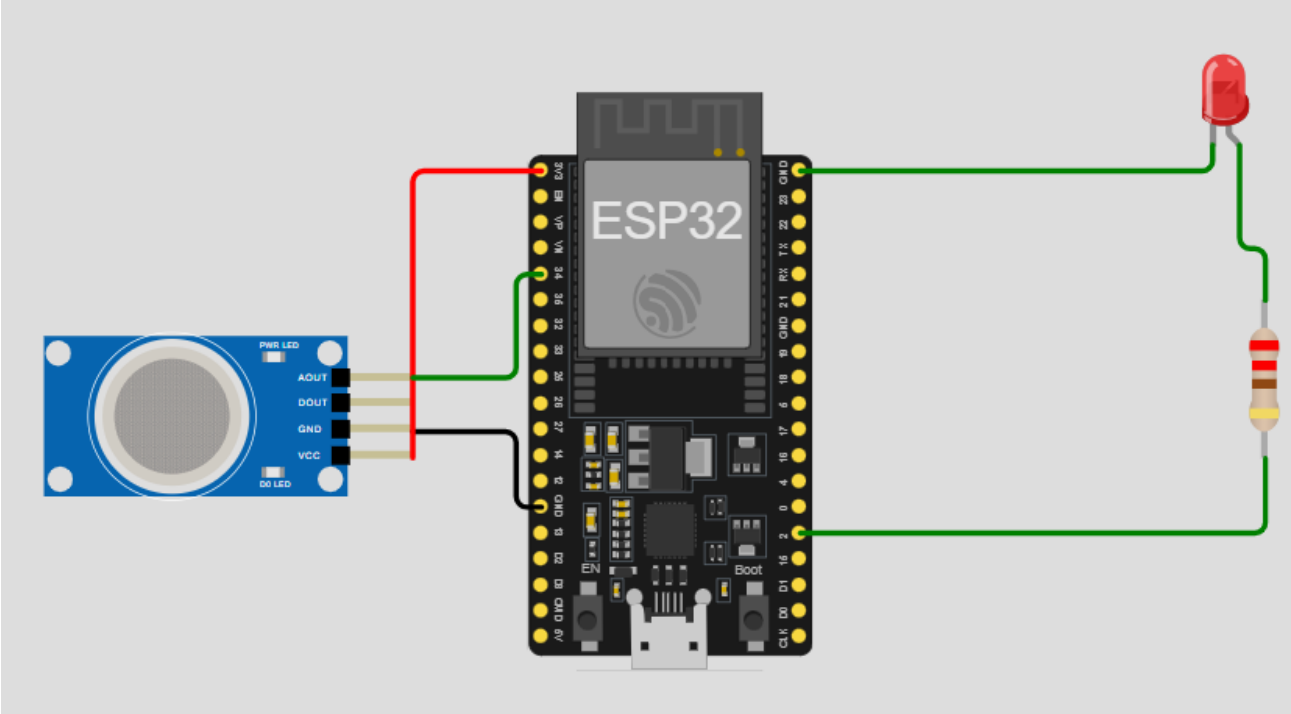
The **ESP32** reads this varying voltage through its **ADC (Analog-to-Digital Converter)** pins.



Gas leakage detection is an important safety application in industrial plants, homes, and laboratories. Harmful gases such as LPG, methane, carbon monoxide, and smoke can cause serious accidents if not detected early. An electronic gas detection system continuously monitors the surrounding air and provides an alert when gas concentration exceeds a safe limit.

It is important to note that MQ gas sensors do not provide precise gas concentration in parts per million (ppm) without calibration. Therefore, the system works on **relative measurement**, making it suitable for gas detection and safety warning applications rather than exact gas quantification.

INTERFACING DIAGRAM:



PROCEDURE:

- Open the editor and create a new ESP32 project.
- Place the ESP32 Dev Module, MQ gas sensor, LED, and resistor on the workspace.
- Connect the MQ sensor to the ESP32 as per the connection table.
- Connect the LED in series with a 220Ω resistor to an ESP32 digital pin.
- Open the Arduino code editor.
- Configure GPIO34 as an analog input and the LED pin as output.
- Read the analog value from the MQ sensor using analogRead(34).
- Display the sensor value on the Serial Monitor.
- Set threshold levels to classify the gas concentration as SAFE, WARNING, or DANGER.
- Turn ON the LED when the gas level crosses the danger threshold.
- Run the simulation and observe the LED and serial output.
- Increase gas concentration in Wokwi and note the change in readings.

PROGRAM:

RESULTS:

ANALYSIS AND INTERFERENCE:

SAMPLE VIVA-VOCE QUESTIONS:

1) What is the Principle of MQ2 Gas sensor?
2) What is the function of a gas sensor?
3) How do you interface a gas sensor?
4) What is the range of a gas sensor?
5) Is MQ2 Gas Sensor is Analog Sensor? If yes then why?